

1) Medical Image Enhancement
Which gray level transformation would you
Recommend?

2) Why is it Suitable?
It increases the contrast between dark
Region without affecting the Brighter
Region too much

3) This makes hidden anatomical details
more visible

4. What effect will it have on the Image?
Dark Lines become clearer.

Bones remain visible with Improved
Contrast

Overall Image Quality and diagnostic
Accuracy Improve.

2. Night Time Surveillance

3: Which Transformation would Improve the
Visibility?

Log Transformation

2. Why is it more Suitable than negative
Transformation

Log Transformation enhances Low Intensity
dark pixels by expanding their gray
levels

It brightens the dark areas effectively
negative transformation only turns bright
dark becomes bright and bright becomes dark
but does not improve contrast in dark regions
it affects the appearance of the
enhanced image.

The parking lot becomes brighter. Cars, people
and objects become easier to identify.
Shadow details become visible.

Overall visibility improves significantly.

3 Fingerprint Recognition

1. Which gray level transformation would you
apply?

Contrast Stretching

2. Explain why.

It increases the contrast between fingerprint
ridges and valleys

fine ridge patterns become sharper and more
distinguishable

This improves the accuracy of fingerprint
matching and recognition.